Besides, for the term $B_2(\theta)$ in Eq. (A.15), we have that

$$
\begin{aligned}
B_2(\theta) &:= \frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau}\Big[\|W_{11}^{KQ} x_{\tau,q}\|_2^2 \cdot \|w_{22}^V Y_\tau^{\text{sfx}}\|_2^2\Big] \\
&= \frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2} \cdot (w_{22}^V)^2 \cdot \mathop{\mathbb{E}}_{\tau}\Big[x_{\tau,q}^\top \cdot (W_{11}^{KQ})^\top W_{11}^{KQ} \cdot x_{\tau,q}\Big] \cdot \mathop{\mathbb{E}}_{\tau}\Big[w_\tau^\top \cdot X_\tau^{\text{sfx}} (X_\tau^{\text{sfx}})^\top \cdot w_\tau\Big] \\
&= \frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2} \cdot (w_{22}^V)^2 \cdot \underbrace{\text{Tr}\Big[(W_{11}^{KQ})^\top W_{11}^{KQ} \cdot \Lambda\Big]}_{\text{by Lemma A.2}} \cdot \mathop{\mathbb{E}}_{\tau}\Big[w_\tau^\top \cdot M_{\text{train}}\Lambda \cdot w_\tau\Big] \\
&= \frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2} \cdot (w_{22}^V)^2 \cdot \underbrace{\text{Tr}\Big[W_{11}^{KQ} \cdot \Lambda \cdot (W_{11}^{KQ})^\top\Big]}_{\text{by Lemma A.3}} \cdot \underbrace{\text{Tr}\Big[M_{\text{train}}\Lambda \cdot I_d\Big]}_{\text{by Lemma A.2}} \\
&= 2\epsilon^2 \cdot \frac{M_{\text{train}}^2 \text{Tr}(\Lambda)}{(N+M_{\text{train}}\Lambda^{\frac{1}{2}})^2} \cdot \text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}}) \cdot \Lambda \cdot (w_{22}^V W_{11}^{KQ})^\top\Big] \\
&= 2\epsilon^2 \cdot \psi(M_{\text{train}}) \cdot \text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}}) \cdot \Lambda \cdot (w_{22}^V W_{11}^{KQ})^\top\Big]. \qquad \text{(A.20)}
\end{aligned}
$$

Finally, by inserting Eqs. (A.19) and (A.20) into Eq. (A.15), we have

$$
\begin{aligned}
&\tilde{\mathcal{L}}^{\text{adv}}(\theta) \\
&:= 2 \cdot \text{Tr}\Big[\Gamma(M_{\text{train}})\Lambda \cdot (w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}}) \cdot (w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})^\top\Big] - 4 \cdot \text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}}) \cdot \Lambda^{\frac{3}{2}}\Big] \\
&\quad + 2 \cdot \text{Tr}(\Lambda) + 2\epsilon^2 \cdot \psi(M_{\text{train}}) \cdot \text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}}) \cdot \Lambda \cdot (w_{22}^V W_{11}^{KQ})^\top\Big] \\
&= 2 \cdot \text{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d) \cdot (w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}}) \cdot (w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})^\top\Big] \\
&\quad - 4 \cdot \text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}}) \cdot \Lambda^{\frac{3}{2}}\Big] + 2 \cdot \text{Tr}(\Lambda),
\end{aligned}
$$

which completes the proof. □

Based on the simplified surrogate AT loss, the closed-form global minimizer θ_* for the surrogate AT problem is then calculated in the following Lemma A.8.

Lemma A.8. *Suppose Assumption 1 holds. Then, $\theta_* := (W_*^V W_*^{KQ})$ is a minimizer for the surrogate AT loss $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ in Eq. (8) if and only if $w_{*,22}^V W_{*,11}^{KQ} = (\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-1}\Lambda$.*

Proof. For the simplified surrogate AT loss proved in Lemma A.7, we rewrite it as follows,

$$
\begin{aligned}
&\tilde{\mathcal{L}}^{\text{adv}}(\theta) \\
&= 2\text{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d) \cdot (w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}}) \cdot (w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})^\top\Big] \\
&\quad - 4\text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}}) \cdot \Lambda^{\frac{3}{2}}\Big] + 2\text{Tr}(\Lambda) \\
&= 2 \cdot \text{Tr}\Big[(\Gamma_{\text{train}}\Lambda + \epsilon^2\psi_{\text{train}}I_d) \cdot \Big(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}} - (\Gamma_{\text{train}}\Lambda + \epsilon^2\psi_{\text{train}}I_d)^{-1}\Lambda^{\frac{3}{2}}\Big) \\
&\qquad\qquad \cdot \Big(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}} - (\Gamma_{\text{train}}\Lambda + \epsilon^2\psi_{\text{train}}I_d)^{-1}\Lambda^{\frac{3}{2}}\Big)^\top\Big] \\
&\quad - \text{Tr}\Big[\Lambda^3(\Gamma_{\text{train}}\Lambda + \epsilon^2\psi_{\text{train}}I_d)^{-1}\Big] + 2 \cdot \text{Tr}(\Lambda), \qquad \text{(A.21)}
\end{aligned}
$$

where $\Gamma_{\text{train}} := \Gamma(M_{\text{train}})$ and $\psi_{\text{train}} := \psi(M_{\text{train}})$.

Notice that the second and third terms in Eq. (A.21) are constants. Besides, the matrix $(\Gamma_{\text{train}}\Lambda + \epsilon^2\psi I_d)$ in the first term in Eq. (A.21) is positive definite, which means that this first term is non-negative. As a result, the surrogate AT loss $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ will be minimized when the first term in Eq. (A.21) becomes zero. This can be achieved by setting

$$
w_{*,22}^V W_{*,11}^{KQ}\Lambda^{\frac{1}{2}} - (\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-1}\Lambda^{\frac{3}{2}} = 0,
$$

which is

$$w_{*,22}^V W_{*,11}^{KQ} = (\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-1}\Lambda.$$

The proof is completed. □

We now turn to prove an PL-inequality for the surrogate AT problem. The proof idea follows that in [69]. Specifically, we will first prove several technical lemmas (*i.e.*, Lemma A.9, Lemma A.10, and Lemma A.11), and then present the PL-inequality in Lemma A.12, which can then enable the surrogate AT model in Eq. (9) approaches its global optimal solution.

Lemma A.9. *Suppose Assumption 1 holds and the model $f_{\text{LSA},\theta}$ is trained via minimizing the surrogate AT loss $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ in Eq. (9) with continuous training flow. Then, for any continuous training time $t \geq 0$, we uniformly have that*

$$(w_{22}^V(t))^2 = \text{Tr}[W_{11}^{KQ}(t)(W_{11}^{KQ}(t))^\top].$$

Proof. Since the model is trained via continuous gradient flow, thus $\partial_t W_{11}^{KQ}(t)$ can be calculated based on the simplified surrogate AT loss proved in Lemma A.7 as follows,

$$\begin{aligned}
&\partial_t W_{11}^{KQ}(t) := -\partial_{W_{11}^{KQ}}\tilde{\mathcal{L}}^{\text{adv}}(\theta) \\
&= -2\cdot\partial_{W_{11}^{KQ}}\text{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})^\top\Big] \\
&\quad + 4\cdot\partial_{W_{11}^{KQ}}\text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot\Lambda^{\frac{3}{2}}\Big] \\
&= -2\cdot(w_{22}^V)^2\cdot\partial_{W_{11}^{KQ}}\text{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\cdot W_{11}^{KQ}\cdot\Lambda\cdot(W_{11}^{KQ})^\top\Big] + \underbrace{4w_{22}^V\Lambda^2}_{\text{by Lemma A.4}} \\
&= \underbrace{-4\cdot(w_{22}^V)^2\cdot(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\cdot W_{11}^{KQ}\cdot\Lambda}_{\text{by Lemma A.4}} + 4w_{22}^V\Lambda^2. \qquad \text{(A.22)}
\end{aligned}$$

Similarly, for $\partial_t w_{22}^V(t)$, we have

$$\begin{aligned}
&\partial_t w_{22}^V(t) := -\partial_{w_{22}^V}\tilde{\mathcal{L}}^{\text{adv}}(\theta) \\
&= -2\cdot\partial_{w_{22}^V}\text{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})^\top\Big] \\
&\quad + 4\cdot\partial_{w_{22}^V}\text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot\Lambda^{\frac{3}{2}}\Big] \\
&= -4w_{22}^V\cdot\text{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\cdot(W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot(W_{11}^{KQ}\Lambda^{\frac{1}{2}})^\top\Big] \\
&\quad + 4\cdot\text{Tr}\Big[(W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot\Lambda^{\frac{3}{2}}\Big]. \qquad \text{(A.23)}
\end{aligned}$$

Combining Eqs (A.22) and (A.23), we thus have

$$\begin{aligned}
&\text{Tr}\Big[\partial_t W_{11}^{KQ}(t)(W_{11}^{KQ}(t))^\top\Big] \\
&= -4\cdot(w_{22}^V)^2\cdot\text{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\cdot(W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot(W_{11}^{KQ}\Lambda^{\frac{1}{2}})^\top\Big] \\
&\quad + 4w_{22}^V\cdot\text{Tr}\Big[\Lambda^{\frac{3}{2}}\cdot(\Lambda^{\frac{1}{2}}W_{11}^{KQ})^\top\Big] \\
&= (\partial_t w_{22}^V(t))w_{22}^V(t),
\end{aligned}$$

which further indicates that

$$\begin{aligned}
&\partial_t\text{Tr}\Big[W_{11}^{KQ}(t)(W_{11}^{KQ}(t))^\top\Big] \\
&= \text{Tr}\Big[\partial_t W_{11}^{KQ}(t)\cdot(W_{11}^{KQ}(t))^\top\Big] + \text{Tr}\Big[W_{11}^{KQ}(t)\cdot\partial_t(W_{11}^{KQ}(t))^\top\Big] \\
&= (\partial_t w_{22}^V(t))\cdot w_{22}^V(t) + W_{22}^V(t)\cdot(\partial_t w_{22}^V(t)) = \partial_t(w_{22}^V(t)^2). \qquad \text{(A.24)}
\end{aligned}$$

Finally, according to Assumption 1, we have that when the continuous training time is $t = 0$,

$$\mathrm{Tr}\Big[W_{11}^{KQ}(0)(W_{11}^{KQ}(0))^\top\Big] = \|W_{11}^{KQ}(0)\|_F^2 = \sigma^2 = w_{22}^V(0)^2.$$

Combine with Eq. (A.24), we thus have that

$$\mathrm{Tr}\Big[W_{11}^{KQ}(t)(W_{11}^{KQ}(t))^\top\Big] = w_{22}^V(t)^2, \quad \forall t \geq 0.$$

The proof is completed. □

Lemma A.10. *Suppose Assumption 1 holds and the model $f_{\mathrm{LSA},\theta}$ is trained via minimizing the surrogate AT loss $\tilde{\mathcal{L}}^{\mathrm{adv}}(\theta)$ in Eq. (9) with continuous training flow. Then, if the parameter σ in Assumption 1 satisfies*

$$\sigma < \sqrt{\frac{2}{d \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2}},$$

we have $w_{22}^V(t) > 0$ holds for any continuous training time $t \geq 0$.

Proof. According to the simplified AT loss calculated in Lemma A.7, we know that if $w_{22}^V(t) = 0$, then $\tilde{\mathcal{L}}^{\mathrm{adv}}(\theta_t) = 2\mathrm{Tr}(\Lambda)$. Besides, under Assumption 1, we have $w_{22}^V(0) = \sigma > 0$. Therefore, if we can show that $\tilde{\mathcal{L}}^{\mathrm{adv}}(\theta_t) \neq 2\mathrm{Tr}(\Lambda)$ for any $t \geq 0$, then it is proved that $w_{22}^V(t) > 0$ for any $t \geq 0$.

To this end, we first analyze the surrogate AT loss $\tilde{\mathcal{L}}^{\mathrm{adv}}(\theta_t)$ at the initial training time $t = 0$. By applying Assumption 1, we have

$$\begin{aligned}
&\tilde{\mathcal{L}}^{\mathrm{adv}}(\theta_0) \\
&= 2\mathrm{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d) \cdot (w_{22}^V(0)W_{11}^{KQ}(0)\Lambda^{\frac{1}{2}}) \cdot (w_{22}^V(0)W_{11}^{KQ}(0)\Lambda^{\frac{1}{2}})^\top\Big] \\
&\quad - 4\mathrm{Tr}\Big[(w_{22}^V(0)W_{11}^{KQ}(0)\Lambda^{\frac{1}{2}}) \cdot \Lambda^{\frac{3}{2}}\Big] + 2\mathrm{Tr}(\Lambda) \\
&= 2\mathrm{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d) \cdot (\sigma^2\Theta\Theta^\top\Lambda^{\frac{1}{2}}) \cdot (\sigma^2\Theta\Theta^\top\Lambda^{\frac{1}{2}})^\top\Big] \\
&\quad - 4\mathrm{Tr}\Big[(\sigma^2\Theta\Theta^\top\Lambda^{\frac{1}{2}}) \cdot \Lambda^{\frac{3}{2}}\Big] + 2\mathrm{Tr}(\Lambda) \\
&= 2\sigma^4 \cdot \mathrm{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1} \cdot \Lambda\Theta\Theta^\top\Lambda\Theta\Theta^\top\Big] - 4\sigma^2\|\Lambda\Theta\|_F^2 + 2\mathrm{Tr}(\Lambda) \\
&\leq 2\sigma^4 \cdot \underbrace{d \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2 \cdot \|\Lambda\Theta\Theta^\top\Lambda\Theta\Theta^\top\|_2}_{\text{by Lemma A.5}} - 4\sigma^2\|\Lambda\Theta\|_F^2 + 2\mathrm{Tr}(\Lambda) \\
&\leq 2\sigma^4 \cdot d \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2 \cdot \|\Lambda\Theta\Theta^\top\Lambda\|_F \cdot \|\Theta\Theta^\top\|_F \\
&\quad - 4\sigma^2\|\Lambda\Theta\|_F^2 + 2\mathrm{Tr}(\Lambda) \\
&\leq 2\sigma^4 \cdot d \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2 \cdot \|\Lambda\Theta\|_F^2 \cdot 1 - 4\sigma^2\|\Lambda\Theta\|_F^2 + 2\mathrm{Tr}(\Lambda) \\
&= 2\sigma^2 \cdot \|\Lambda\Theta\|_F^2 \cdot (d \cdot \sigma^2 \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2 - 2) + 2\mathrm{Tr}(\Lambda). \qquad \text{(A.25)}
\end{aligned}$$

By Assumption 1, we have $\|\Lambda\Theta\|_F^2 > 0$. Thus, when $(d \cdot \sigma^2 \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2 - 2) < 0$, which is

$$\sigma < \sqrt{\frac{2}{d \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2}},$$

we will have $\tilde{\mathcal{L}}^{\mathrm{adv}}(\theta_0) < \mathrm{Tr}(\Lambda)$.

Finally, since the surrogate AT loss $\tilde{\mathcal{L}}^{\mathrm{adv}}(\theta_t)$ is minimized with continuous gradient, thus when the above condition holds, for any $t > 0$, we always have that $\tilde{\mathcal{L}}^{\mathrm{adv}}(\theta_t) \leq \tilde{\mathcal{L}}^{\mathrm{adv}}(\theta_0) < \mathrm{Tr}(\Lambda)$.

The proof is completed. □